

Parking lot - LLD

Design

client → main

Pricing

Bike Hourly
car hourly
surge charge weekend

{ strategy }

pricing should be automatically applied based on vehicle

~~slots~~ / spots

compact ✓

large ✓

Heavy → ✓

ISpot

getsize()

isavailable

assignVehicle()

setstate()

State

available ✓

occupied

Reserved

assign()

release()

Singleton

vehicles

getsize()

4wheeler

2wheeler

heavy 4wheeler

Ticket

Vehicle ✓

Entry ✓

Exit →

spot ✓

Parking Lot

List<Floor> ✓

{ Singleton }

add floor

get floors()

getAvailablespots()

Floor

List<spot> ✓

floor name

addspot()

getspots()

getAvailablespots()



Composite pattern

How client communicates with our system.

class Parking facade {

Ticket park(vehicle)

price

double unpark(Ticket) {

spot finder {

get Available spot(vehicle) {